

Reconstructing the invisible: The use of geophysics in medical imaging

By Tiffany Ajayi and Mathilde Johnsen

Main topics

- Geophysical-inspired methods applied to medicine
- Methods used to image the brain and in high contrast areas
- Acquisition geometry
- Results
- Discussion
- Conclusion

Geophysical-inspired methods applied in medical imaging

Table 1: Different techniques in medicine that have their roots in geophysical principles and their application

Technique in medicine	Geophysical principle	Application
Electrical impedance tomography (EIT)	Electrical techniques, changes in electrical conductivity, track alterations in electrical conductivity within the body.	Lung imaging, brain and breast imaging.
Elastography	Seismic waves move through the subsurface of the earth and interact with various layers to disclose details. In medicine, it measures the movement of mechanical waves across the body to determine the stiffness of the tissue.	Cancer detection, breast imaging, tissue health, and identifying anomalies.
MRI	The concept of a magnetic field and signal detection. Geophysicists care about how different materials react to magnetic signals, while medical imaging use it to highlight variations in tissue composition.	Disease detection, diagnosis, and treatment monitoring
Ultrasound	High-frequency sound waves are used to produce images of soft tissue and organs. These techniques were inspired by seismic approaches, where geophysicists used the seismic waves to study the Earth’s subsurface.	Obstetrics and gynaecology, vascular imaging and breast imaging.

Traditional medical imaging

MRI	Ultrasound
Expensive Traditional tomography Motion sensitive	Wavefront distortions Reflection, refraction etc. Limited by poor image quality

Methods – Equations

Step	Equations	Parametres
1) Forward Modelling of Wave Propagation	$\mathbf{A}u = \mathbf{s}$	$\mathbf{A}$ – wave operator u – simulated wavefield $\mathbf{s}$ –the source term
2) High-frequency approximation based on the eikonal equation	$\frac{d}{ds} \left(n \frac{d\mathbf{r}}{ds} \right) = \nabla n$	n – refractive index c – speed of sound
3) Model-based data-fitting approach	$f(m) = \frac{1}{2} (\mathbf{p} - \mathbf{d})^T (\mathbf{p} - \mathbf{d})$	$\mathbf{d}$ – observed data $\mathbf{p}$ – synthetic $(\mathbf{p} = \mathbf{R} \cdot \mathbf{u})$
4) Inverse-scattering based reconstruction	$I_{ijk} = \alpha(\text{SoS})_{ijk} + \beta(\text{Atten})_{ijk} + \gamma(\text{Refl})_{ijk}$	I_{ijk} – fused image intensity at voxel (i,j,k) $\alpha(\text{SoS})_{ijk}$ – Speed of sound $\beta(\text{Atten})_{ijk}$ – Attenuation $\gamma(\text{Refl})_{ijk}$ – Reflection

Acquisition geometry in brain imaging

- Yellow circle → source transductor
- Red ellipse → array location
- Side length of 256 mm
- 1024-point sources and receivers
- Full angular coverage and high sensitivity

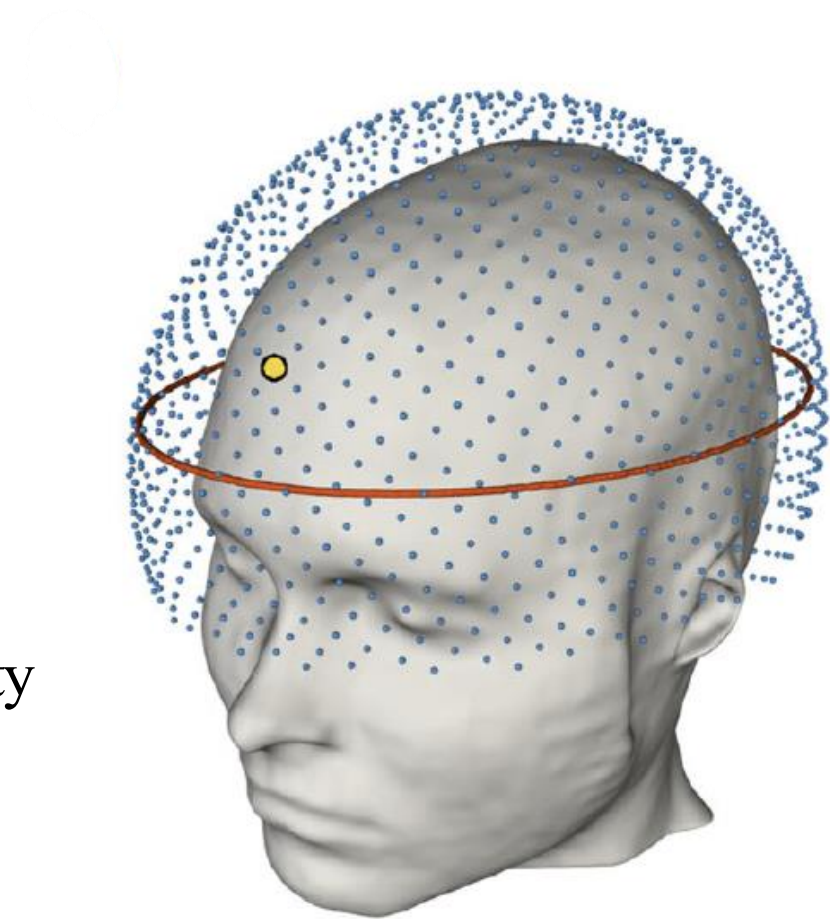


Figure 1: Illustrates the acquisition geometry for brain imaging

Acquisition in bone imaging

- 2048 individual ray elements
- Fully 3D data
- 360 degrees shooting pseudo plane every 2°
- 192 beam-formed signals
- 0.35-1.3 MHz

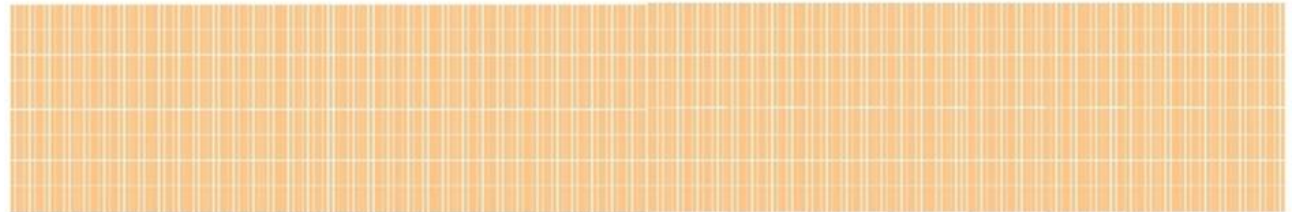
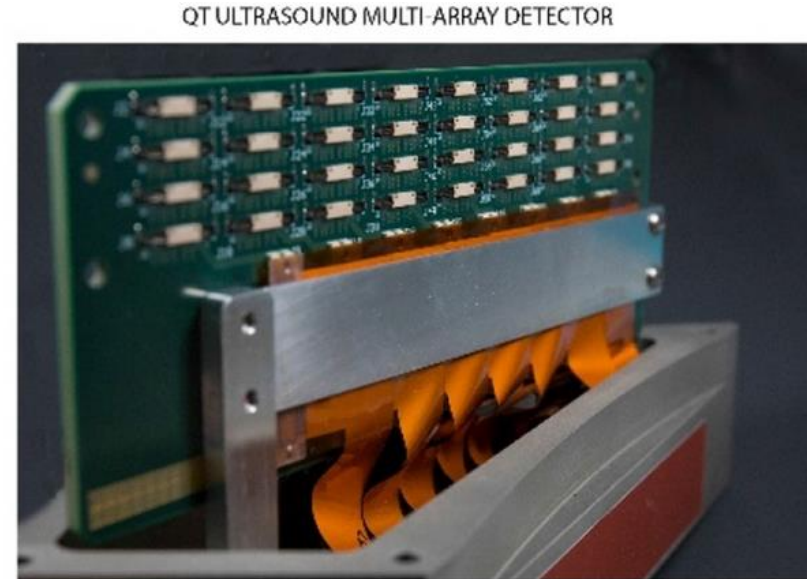


Figure 2: Close-up view of the receiver array used in data acquisition (Wiskin et al., 2020).

Methods - velocities

- Estimate of elastic velocities
 - Spans 100 to 850 kHz
 - 1.6 mm in soft tissue
 - 28 mm in bone
-
- SOS using QTUS – 3D quantitative transmission ultrasound
 - High quantitative accuracy

Table 2: Velocities values for different tissue from brain Imaging (Guasch et al.,2020)

Tissue Types	Acoustic Velocity (m/s)	Variation
Fat	1400	$\pm 7\%$ (≈ 1302 – 1498 m/s)
Muscle tissue & Cartilage	1600	$\pm 7\%$ (≈ 1488 – 1712 m/s)
Cancellous bone	2100	$\pm 14\%$ (≈ 1806 – 2394 m/s)
Cortical bone	2800	$\pm 14\%$ (≈ 2408 – 3192 m/s)
Mandible & vertebrae	3500	

Table 3: Comparison of SOS for muscle fat and cartilage (Wiskin et al., 2020)

Tissue	SOS QTUS 3D (SD) (m/s)	SOS (SD) (m/s)
Vastus medialis	1573 (25.9)	1588.4 (21.6)
Fresh muscle	1572.6 (14.6)	1588.4 (21.6)
Fat	1438.7 (20.5)	1440 (21.4)
Cartilage	1655.4 (14.6)	1660

Results – 3D reconstruction brain imaging

- FWI shows millimetre-scale resolution compared to conventional tomography
- AWI addresses the cycle skipping
- Combining AWI and FWI
- Accurate, high-resolution brain imaging

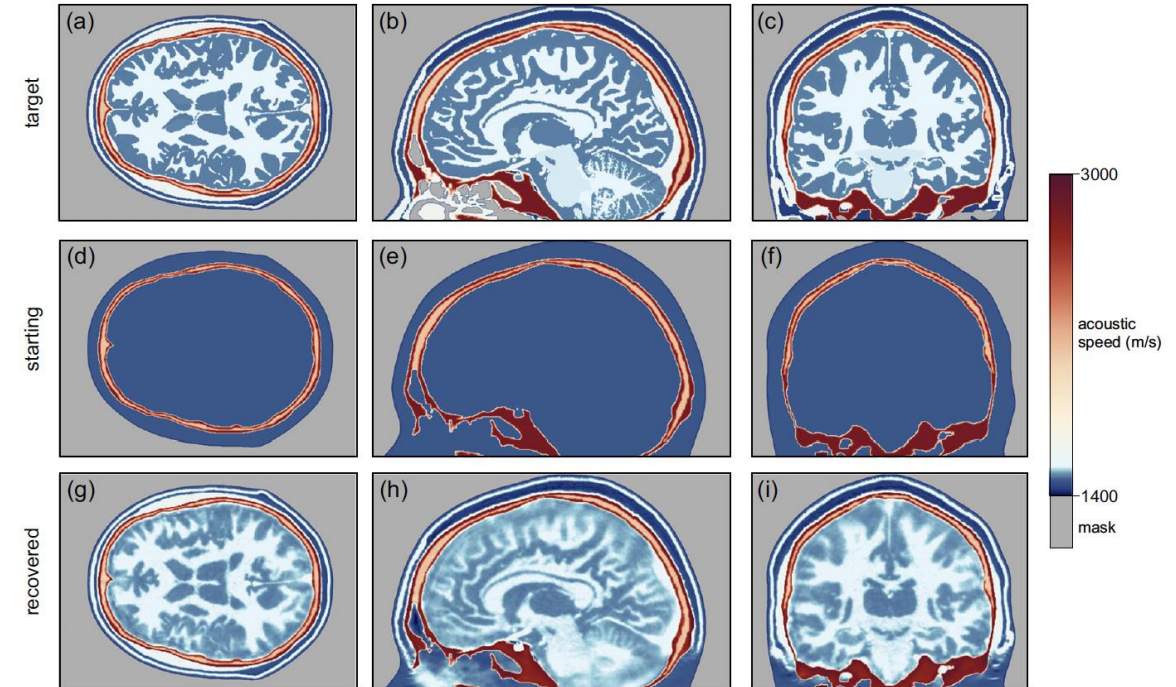


Figure 4: Transverse (left), sagittal (center), and coronal (right) sections. First row is the true model, the second row is the starting and the bottom row is the recovered model. Different acoustic speed and where is located is indicated as a gradient from blue (low) to red (high) (Guasch et al., 2020).

Results – reconstruction of haemorrhage in brain imaging

- FWI's clinical potential
- Is successfully reconstructed and precise
- Not only theoretically powerful but also clinically relevant

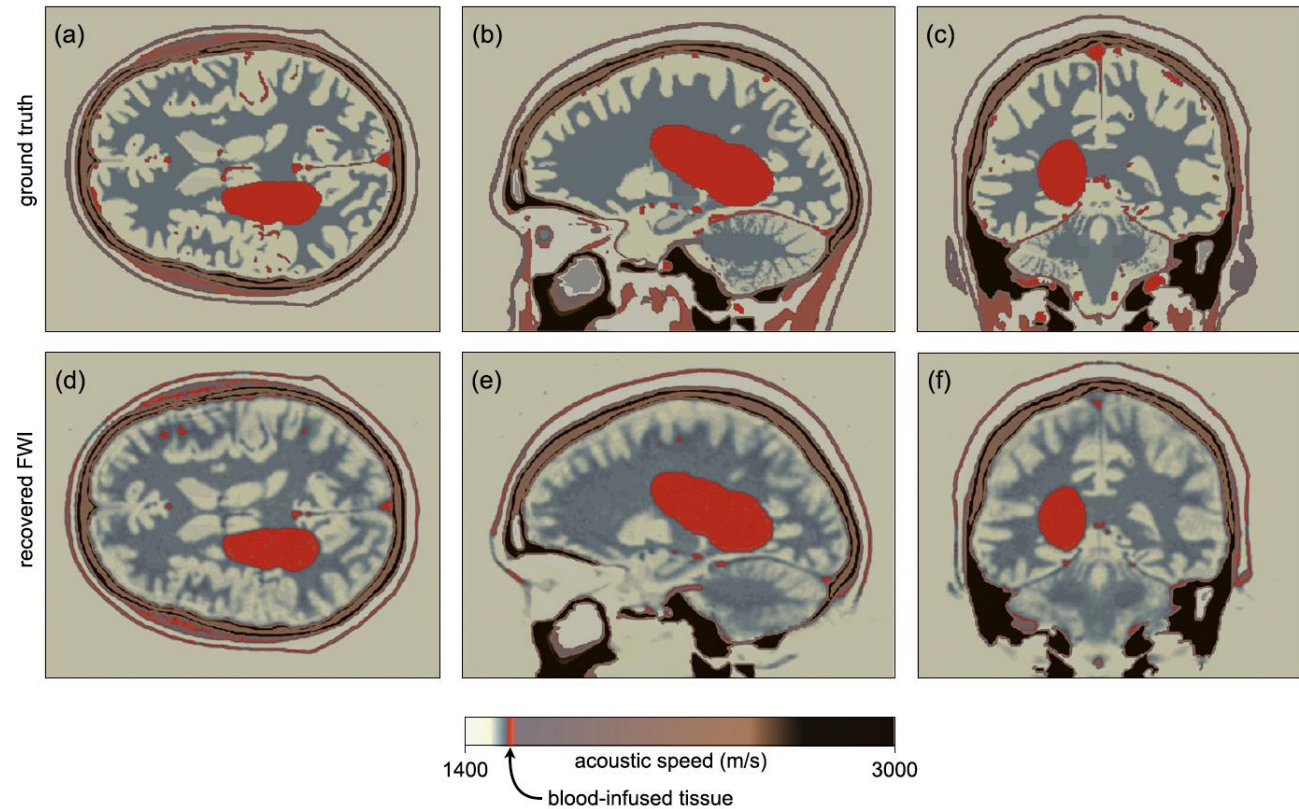


Figure 5: (a)-(c) Slice through a 3D-wave-speed model disturbed by a large haemorrhage, (d)-(f) The same slice THROUGH the FWI. The haemorrhage is well recovered by FWI at high resolution in all slices. The colour scale has been modified to highlight the haemorrhage (Guasch et al., 2020).

Results – High contrast areas

- 3D full-wave, nonlinear inverse scattering algorithm
- High speed tissues
- Brighter area
- Poorly defined

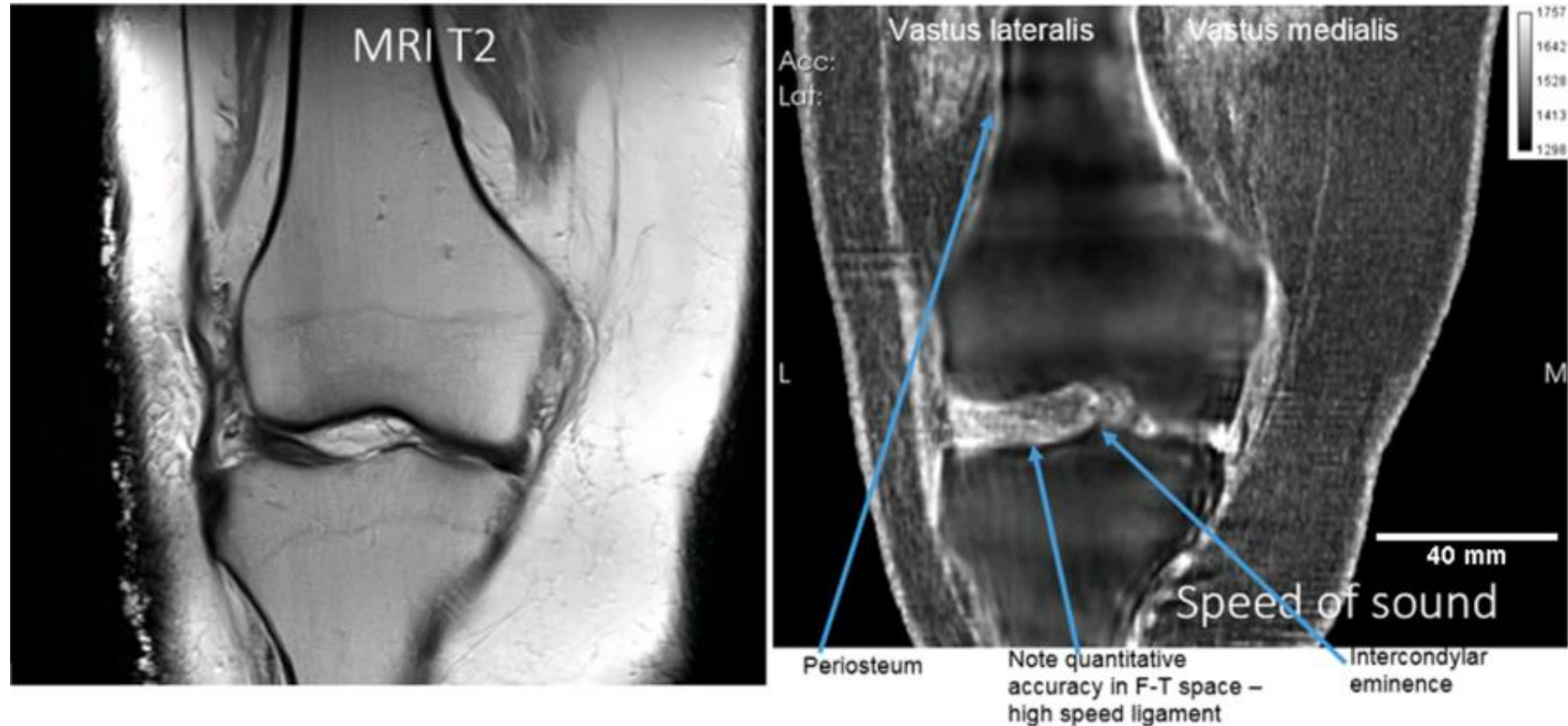


Figure 6: MRI image to the left, and QTUS (quantitative ultrasound) SOS image to the right (Wiskin et al., 2020).

Results – High contrast area

- Correctly visualize high SOS values
- Strong signal
- Distinguish soft tissue structures
- At/below mm scale

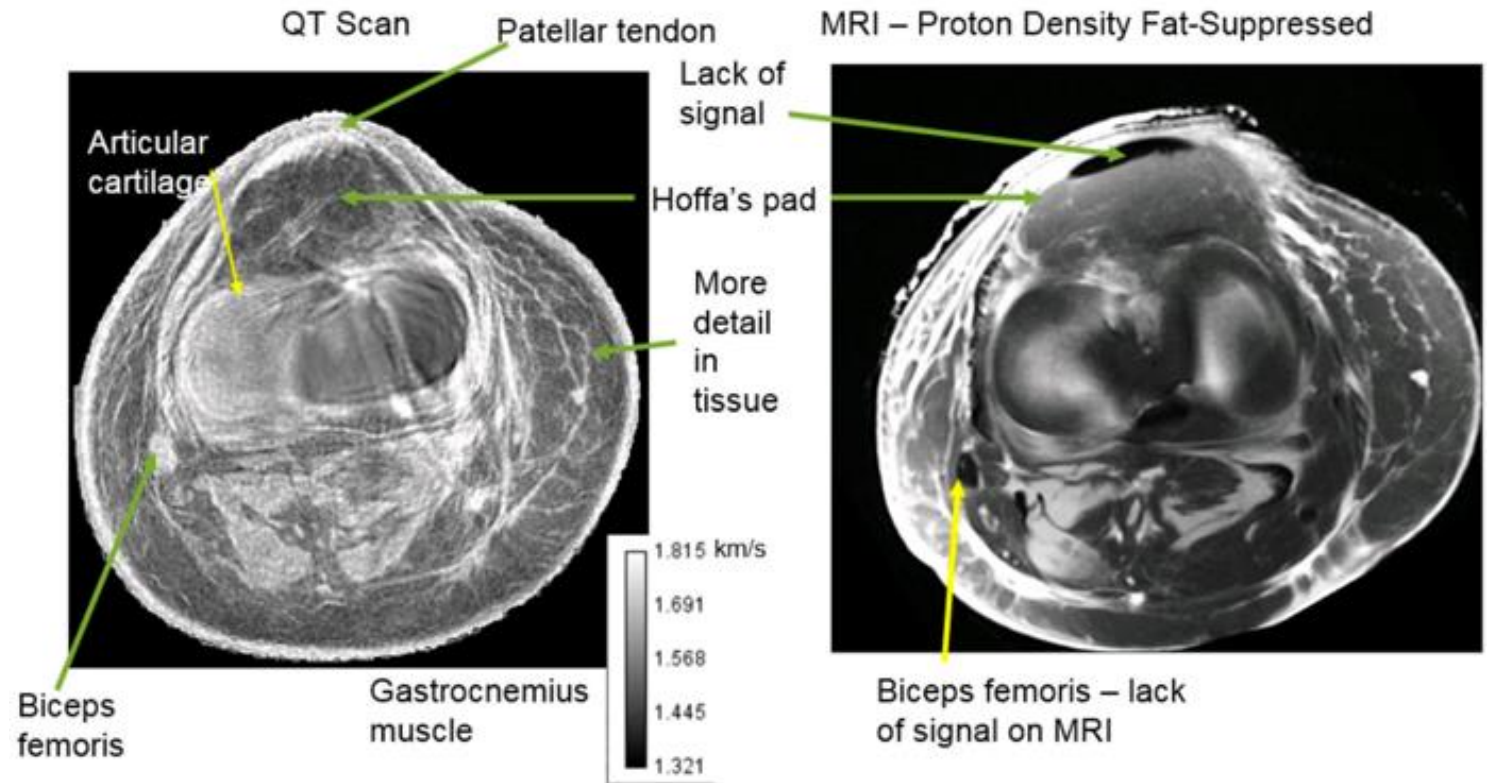


Figure 7: QT scan vs MRI of the gastrocnemius muscle (Wiskin et al., 2020)

Discussion

Challenges

- Poor computational efficiency –30 hours of computation
- Needs to be 200 times faster for emergency
- Reconstruction time reduced to 24 minutes
- Ex vivo experiment

Conclusion

- Complex geometry and strong impedance contrast
- Further technological innovations
- Robust inversion algorithms, improved wave-behavior modelling and hardware systems
- lower mortality rate

References

- Wiskin, J., Malik, B., Borup, D., Pirshafiey, N., & Klock, J. Full wave 3D inverse scattering transmission ultrasound tomography in the presence of high contrast. *Sci Rep* **10**, 20166 (2020). <https://doi.org/10.1038/s41598-020-76754-3>
- Guasch, L., Calderón Agudo, O., Tang, M.-X. Nachev, P., & Warner, M. Full-waveform inversion imaging of the human brain. *npj Digit. Med.* **3**, 28 (2020). <https://doi.org/10.1038/s41746-020-0240-8>
- Singh, R. K., Nayak, N. P., Behl, T., Arora, R., Anwer, M. K., Gulati, M., et al. (2024). Exploring the Intersection of Geophysics and Diagnostic Imaging in the Health Sciences. *Diagnostics*, *14*(2), 139. <https://doi.org/10.3390/diagnostics14020139>